

Variables

Introduction to Programming

StayAhead Training; Sep, 2024

Introduction

Variables

- **Variable:** a name for some data stored in memory
- Data must exist in memory to:
 - write to an output device/disk
 - do some arithmetic
 - perform a comparison
 - make a decision

Variable Declaration and Initialisation

Variables

- Pseudocode, e.g.:
 - SET MyNumber to 42
- Pseudocode keywords are often capitalised
- Some prefer to use = instead of the word to
- Name variables as you see fit, but be consistent

Variable Declaration and Initialisation

Variables

- PY: `my_number = 42`
- _____
- JS: `var myNumber = 42;`
- _____
- JV: `int myNumber = 42;`
- _____
- C#: `int myNumber = 42;`

Data Types

Variables

- Fundamental data types:
 - Boolean (true or false)
 - Number
 - String (text)

Data Types

Variables

- PY: `my_boolean = True`
- _____
- JS: `var myBoolean = true;`
- _____
- JV: `boolean myBoolean = true;`
- _____
- C#: `bool myBoolean = true;`

Data Types

Variables

- PY: `my_number = 42`
- _____
- JS: `var myNumber = 42;`
- _____
- JV: `int myNumber = 42;`
- _____
- C#: `int myNumber = 42;`

Data Types

Variables

- PY: `my_string = "Hello"`
- _____
- JS: `var myString = "Hello";`
- _____
- JV: `String myString = "Hello";`
- _____
- C#: `string myString = "Hello";`

Literals

Variables

- **Literal:** a data value
- `true` is a boolean literal
- `42` is a number (integer) literal
- `"Hello"` is a string literal

Variable Reassignment

Variables

- PY: my_number = 43
- _ _ _ _ _
- JS: myNumber = 43;
- _ _ _ _ _
- JV: myNumber = 43;
- _ _ _ _ _
- C#: myNumber = 43;

Constants

Variables

- **Constant:** a name for some data that cannot be reassigned
- A constant is NOT a variable

Constants

Variables

- JS: `const MY_CONSTANT = 0.2;`
- -----
- JV: `final float MY_CONSTANT = 0.2f;`
- -----
- C#: `const float MyConstant = 0.2f;`

Reading From Stdin

Variables

- Pseudocode, e.g.:
 - INPUT MyNumber from stdin
- There are two operations here:
 1. Input from stdin
 2. Storage of the input value in memory

Reading From Stdin

Variables

- PY: `data = input()`
- -----
- JS: `var data = prompt();`
- -----
- JV: `Scanner scanner = new Scanner(System.in);`
- JV: `String data = scanner.nextLine();`
- -----
- C#: `string data = Console.ReadLine();`

Writing to Stdout

Variables

- Pseudocode, e.g.:
 - OUTPUT MyNumber to stdout

Writing to Stdout

Variables

- PY: `print(my_string)`
- _____
- JS: `console.log(myString);`
- _____
- JV: `System.out.println(myString);`
- _____
- C#: `Console.WriteLine(myString);`

Writing to Stdout

Variables

- PY: `print("Hello")`
- _____
- JS: `console.log("Hello");`
- _____
- JV: `System.out.println("Hello");`
- _____
- C#: `Console.WriteLine("Hello");`